

Bugs found: 5

1. Array out-of-bounds access: loop iterates i from 0 to 5 (inclusive), but `arr` is declared with size 5 (valid indices 0-4), causing undefined behavior on $i=5$.
2. Integer overflow: `arr[i] = i * 1000000` can exceed the range of `int` (e.g., $5 * 1000000 = 5000000$, and with out-of-bounds $i=5$, 5000000 fits but the write itself is out-of-bounds regardless).
3. Missing `#include <string.h>`: `strcpy` is used without including its header, causing an implicit declaration or compilation error.
4. Buffer too small for `strcpy`: `copy` is declared as `char[4]` but `s` is "Test" which requires 5 bytes (4 chars + null terminator), causing a buffer overflow.
5. Null pointer dereference: `ptr` is explicitly set to `NULL` and then dereferenced with `*ptr`, causing undefined behavior (likely a segfault).